

Finding Likely Term-Deposit Subscribers Under Class Imbalance in a Machine Learning Study of Bank Telemarketing

Chaoran Chen, Haoyang Liu, Haonan Li

Abstract

This project studies whether bank clients will subscribe to a term deposit after a marketing contact. Banks face a practical targeting problem in which wasted calls reduce campaign efficiency and missed likely subscribers reduce potential revenue. The task is statistically difficult, with only 4,640 of 41,188 observations, or 11.27%, belonging to the positive subscription class. We removed call duration to avoid leakage, retained encoded unknown categories, transformed the previous-contact field, then evaluated models with a stratified 80/20 train/test split plus five-fold cross-validation. The final comparison covered classical logistic regression, LASSO logistic regression, Naive Bayes, Random Forest, Weighted XGBoost, and PCA followed by logistic regression. Accuracy is misleading under severe imbalance, so model selection emphasized PR-AUC, F1-score, recall, precision, and cross-validated threshold tuning. Weighted XGBoost gave the strongest final balance on the held-out test set, with an F1 of 0.5236, a PR-AUC of 0.5009, a ROC-AUC of 0.8092, precision of 0.4663, and recall of 0.5970 at a cross-validated threshold of 0.65. Random Forest was a close second and offered slightly higher precision, with logistic and LASSO models serving as useful interpretable benchmarks. The most important practical signals were macroeconomic conditions, prior campaign success, previous contact status, contact channel, and campaign timing.

Introduction and Motivation

Retail banks frequently use direct marketing campaigns to offer term-deposit products. A campaign is expensive when calls are directed to clients with little interest, and missed likely subscribers reduce potential revenue. The project objective is a targeting rule that ranks likely subscribers and preserves a reasonable precision-recall trade-off for a realistic marketing workflow.

The data come from the Bank Marketing dataset in the UCI Machine Learning Repository, originally studied by Moro, Cortez, and Rita (2014). The full file contains 41,188 observations, 20 input variables, and the binary response y , indicating whether the client subscribed to a term deposit. Predictors include client demographics, contact channel and month, current and previous campaign history, and macroeconomic indicators such as employment variation rate, consumer price index, consumer confidence index, Euribor three-month rate, and number of employees.

The analysis is guided by several research questions. Which methods best identify subscribers when the positive class is rare? Do tree-based ensemble models improve over logistic and LASSO regression? Does a generative method or PCA-based model provide a useful benchmark? Can class weighting and threshold tuning improve recall and F1-score? Which predictors produce interpretable marketing insights?

Data Exploration and Preprocessing

The raw dataset contains 41,188 observations and 21 columns, including the binary response variable y . Although there are no literal NA values, several categorical variables use the label unknown as an encoded missing category. Rather than deleting these records or imputing new values, we retained unknown as a valid categorical level because missingness may itself contain useful operational information in a banking context. Table 1 summarizes the dataset after preprocessing.

The variable duration was removed before modeling because it records the last contact duration in seconds and is known only after a call occurs. Including this field would leak post-contact information into a model intended to predict subscription before outreach. We also transformed pdays. In the raw file, the value 999 marks clients with no previous contact rather than an actual day count. We separated this field into pdays_contacted, which records whether a previous contact occurred, and pdays_days, which stores zero for clients without prior contact and the original day count

otherwise. The original pdays variable was then removed. Categorical predictors were converted to factors and encoded as dummy variables during model fitting. Numeric predictors were centered and scaled for models that used preprocessing, including logistic regression, LASSO, Naive Bayes, and PCA-based logistic regression. For the PCA-based logistic model, PCA was applied only to the numeric predictors, while categorical predictors were dummy-coded after the PCA step.

Item	Value
Original observations	41,188
Original columns including response	21
Modeling predictors after preprocessing	20
Positive class count	4,640
Positive class rate	11.27%

Table 1. Dataset summary after preprocessing.

Exploratory analysis showed clear differences across categorical predictors, as shown in Figure 1. Previous campaign information was especially informative. Clients previously contacted had a subscription rate of 63.83%, compared with 9.26% for clients not previously contacted. A previous campaign outcome of success had a rate of 65.11%. Contact channel and timing also mattered. Cellular contacts had a subscription rate of 14.74%, compared with 5.23% for telephone contacts, while March, December, September, and October were well above the overall baseline.

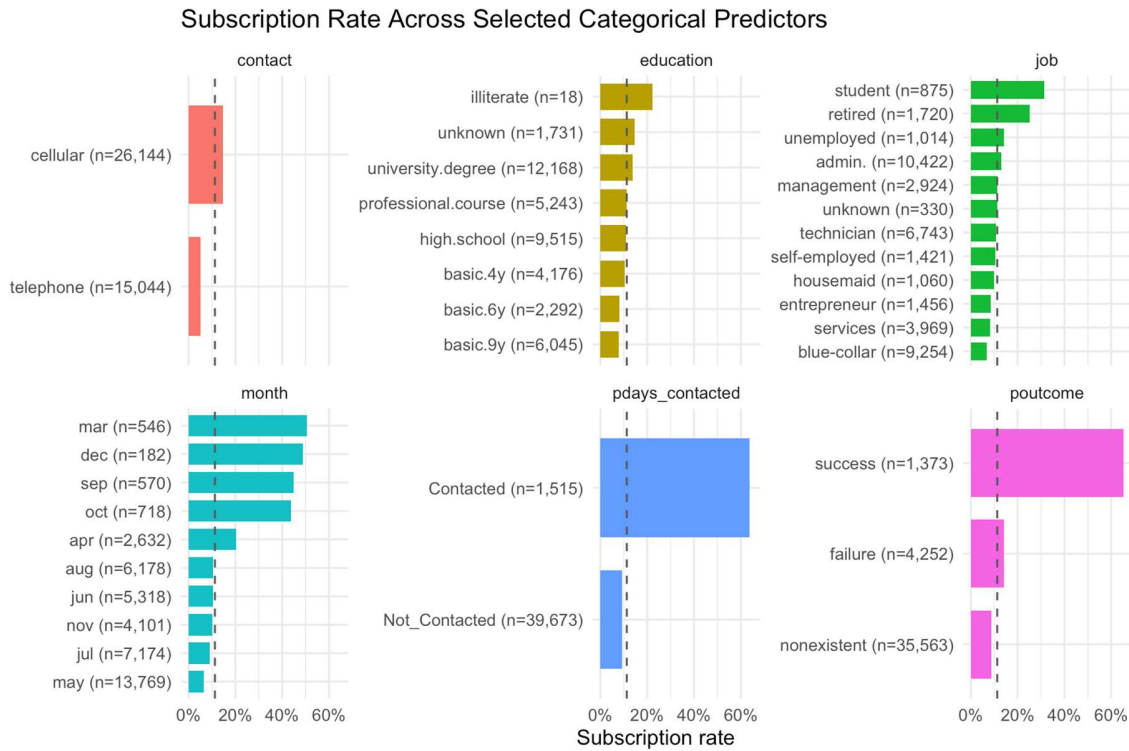


Figure 1. Subscription rates across selected categorical predictors. The dashed line marks the overall positive-class baseline.

Figure 2 extends this pattern to the macroeconomic variables by grouping observations into quintiles. Subscription rates are highest in the lowest quintiles of emp.var.rate, euribor3m, and nr.employed, then decline sharply as these indicators move into stronger economic ranges. This pattern supports the earlier numeric comparison in which subscribers tended to appear under weaker macroeconomic conditions. Figure 3 shows why regularization, tree ensembles, and PCA are useful for the numeric predictors. The macroeconomic indicators carry highly redundant information, with correlations of 0.972 between emp.var.rate and euribor3m, 0.945 between euribor3m and nr.employed, and 0.907 between emp.var.rate and nr.employed.

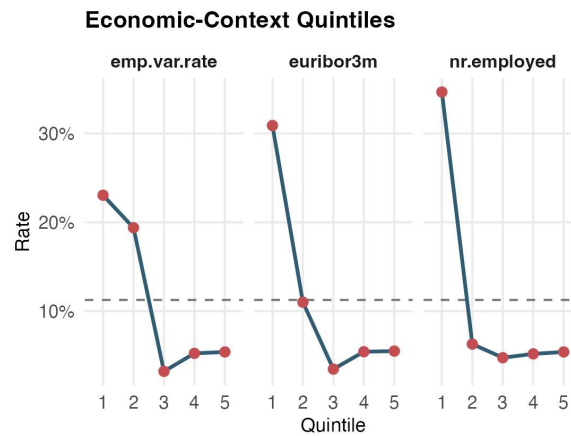


Figure 2. Subscription rates by quintile for *emp.var.rate*, *euribor3m*, and *nr.employed*. The dashed line marks the overall positive-class baseline.

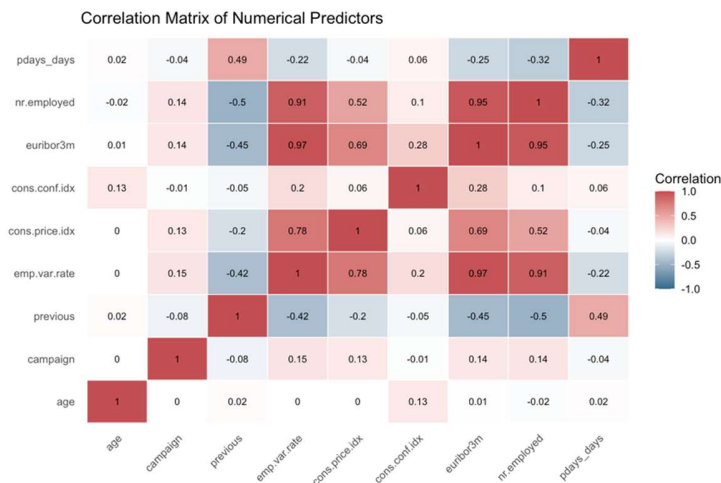


Figure 3. Correlation matrix for numeric predictors. Red cells indicate positive correlation, blue cells indicate negative correlation, and deeper color indicates stronger association.

Modeling, Cross-Validation, and Threshold Selection

The modeling stage used a stratified 80/20 train/test split so that the rare subscriber class kept the same prevalence in both parts of the data. The training set had 29,239 non-subscribers and 3,712 subscribers. The held-out test set had 7,309 non-subscribers and 928 subscribers. All model tuning was performed inside the training set using five-fold cross-validation. In each round, four folds trained the model and the remaining fold produced validation probabilities. This gave out-of-fold predictions for model comparison and threshold selection without looking at the final test set.

Split	No	Yes	Purpose
Training	29,239	3,712	Fit models, run five-fold cross-validation, tune model complexity, and choose classification thresholds.
Test	7,309	928	Evaluate final models only after all tuning choices were made.

Table 2. Stratified split used for model tuning and held-out evaluation.

The same cross-validation folds were used across models, so the comparison would reflect model behavior rather than a different validation split. PR-AUC was emphasized during tuning since it evaluates how well a model retrieves the rare positive class across probability rankings. Accuracy was excluded from selection because a model can appear accurate by predicting almost every client as a non-subscriber.

Logistic Regression and LASSO Logistic Regression

Logistic regression models the log-odds of subscription as an additive function of the predictors. This structure gives a stable classical baseline for comparison with more flexible methods. No complexity grid was needed. The model used the common preprocessing pipeline, class weights, and the same five validation folds.

LASSO logistic regression adds an L1 penalty to the logistic likelihood, shrinking weak effects toward zero and encouraging a sparser linear model. The penalty strength was searched on a logarithmic scale within the five-fold validation design, and validation PR-AUC determined the preferred shrinkage level. The best validation result came from a weak penalty. Stronger penalties removed too much signal, which is visible in the sharp PR-AUC decline in Figure 4.

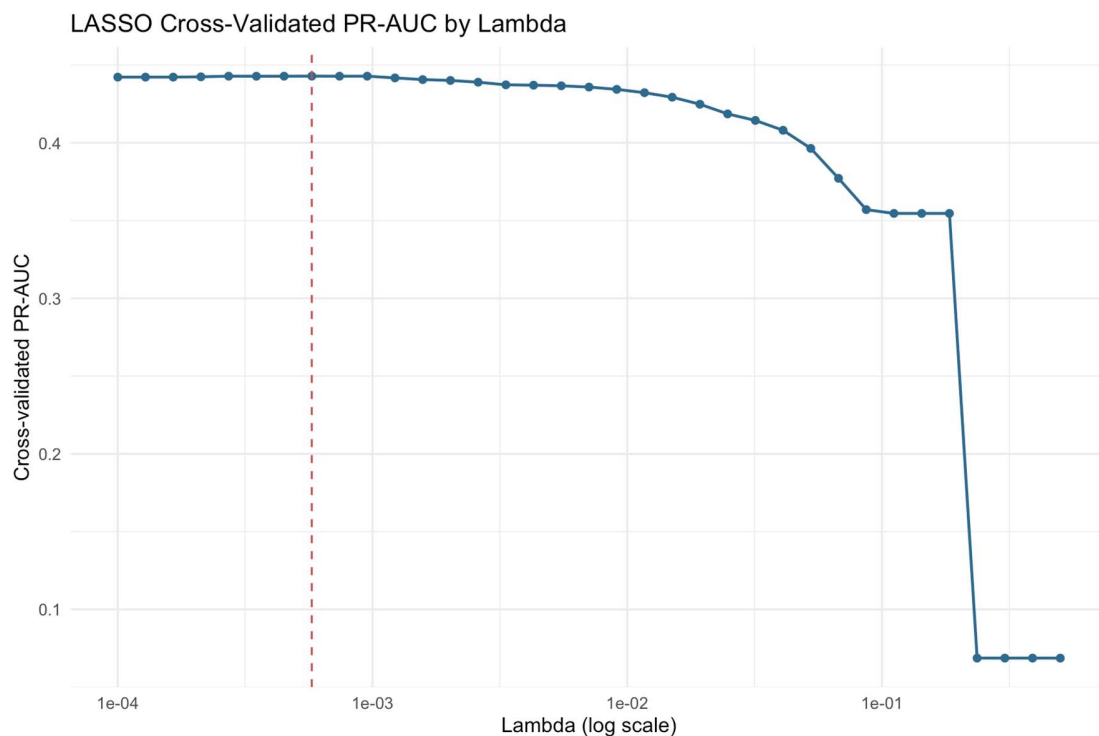


Figure 4. LASSO cross-validated PR-AUC across candidate penalty strengths. The dashed line marks the selected penalty.

Naive Bayes

Naive Bayes estimates class-conditional predictor distributions and combines them through Bayes' rule. The method assumes conditional independence among predictors, making it a useful generative benchmark for a dataset with many categorical indicators. The tuning comparison considered whether Laplace smoothing should adjust sparse categorical patterns. Smoothed and unsmoothed variants produced the same validation PR-AUC, leaving Naive Bayes well below the linear and tree-based alternatives.

Random Forest and Weighted XGBoost

Random Forest aggregates many decision trees built from resampled training data. Random predictor subsets at each split reduce correlation among trees, and the ensemble can capture nonlinearities without a single global linear boundary. Tuning compared the amount of split-level predictor sampling and the minimum leaf size using five-fold PR-AUC. The strongest forest used the most restrictive predictor sampling with a larger terminal-node requirement, yielding a CV PR-AUC of 0.4593. This result favored a conservative forest over a more flexible tree ensemble.

Weighted XGBoost builds trees sequentially, with each new tree correcting errors from the current boosted ensemble. Class weighting increased the influence of subscriber cases during fitting. The tuning search compared sampled combinations of tree count, depth, learning rate, row and column subsampling, child-node constraints, and regularization under the same five-fold validation framework. Several boosted candidates clustered tightly near the top. The selected

candidate achieved a CV PR-AUC of 0.4565 and a CV ROC-AUC of 0.8004, with ROC-AUC used only as a secondary tie-breaker.

PCA Followed by Logistic Regression

PCA summarizes standardized numeric predictors through orthogonal components that retain major directions of variation. The follow-up logistic model tests whether a compressed numeric representation can preserve useful signal after reducing redundancy from correlated economic variables. Categorical predictors were not included in PCA; they were dummy-coded separately for the logistic regression. The preprocessing step chose enough numeric components to reach the 95% cumulative variance target, which required 7 components from 9 numeric predictors. The resulting CV PR-AUC was 0.4364, indicating a useful benchmark that remained below the ensemble models.

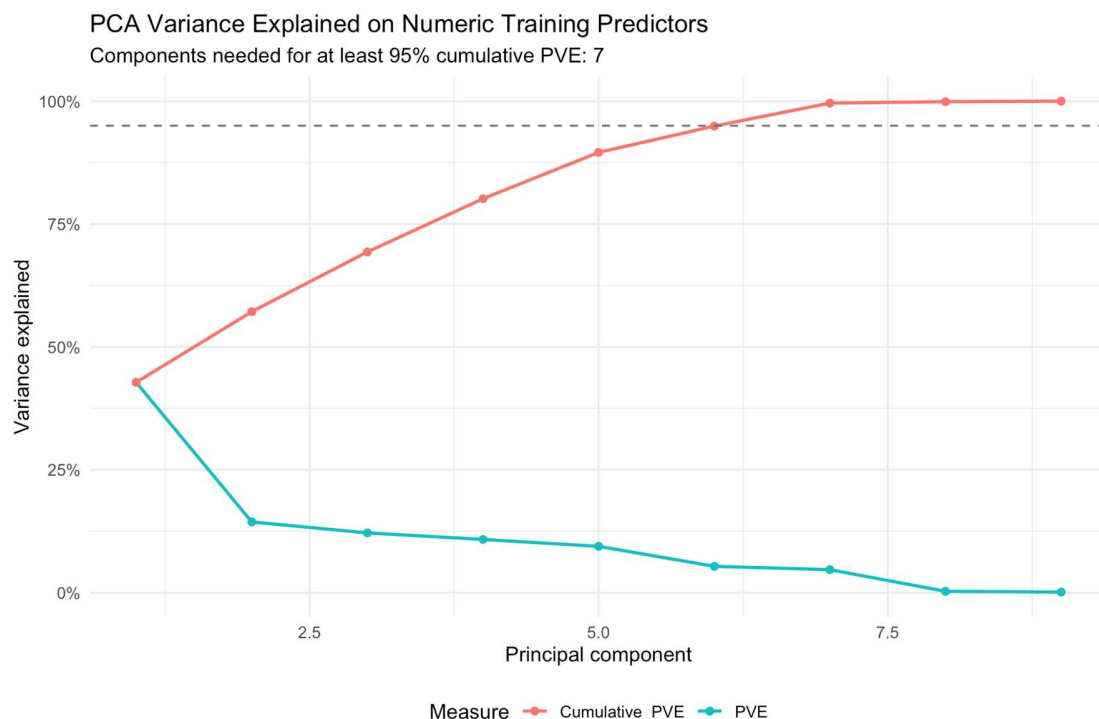


Figure 5. PCA variance explained on numeric training predictors.

Threshold Selection

After tuning model complexity, each model still needed a probability threshold for converting predicted subscription probabilities into Yes/No classifications. The threshold was not chosen from the test set. Instead, the out-of-fold probabilities from cross-validation were evaluated across many possible cutoffs. The selected cutoff for each model was the one that maximized cross-validated F1, with recall used as a secondary preference when F1 was essentially tied. This criterion matches the business problem by balancing unnecessary calls against missed likely subscribers.

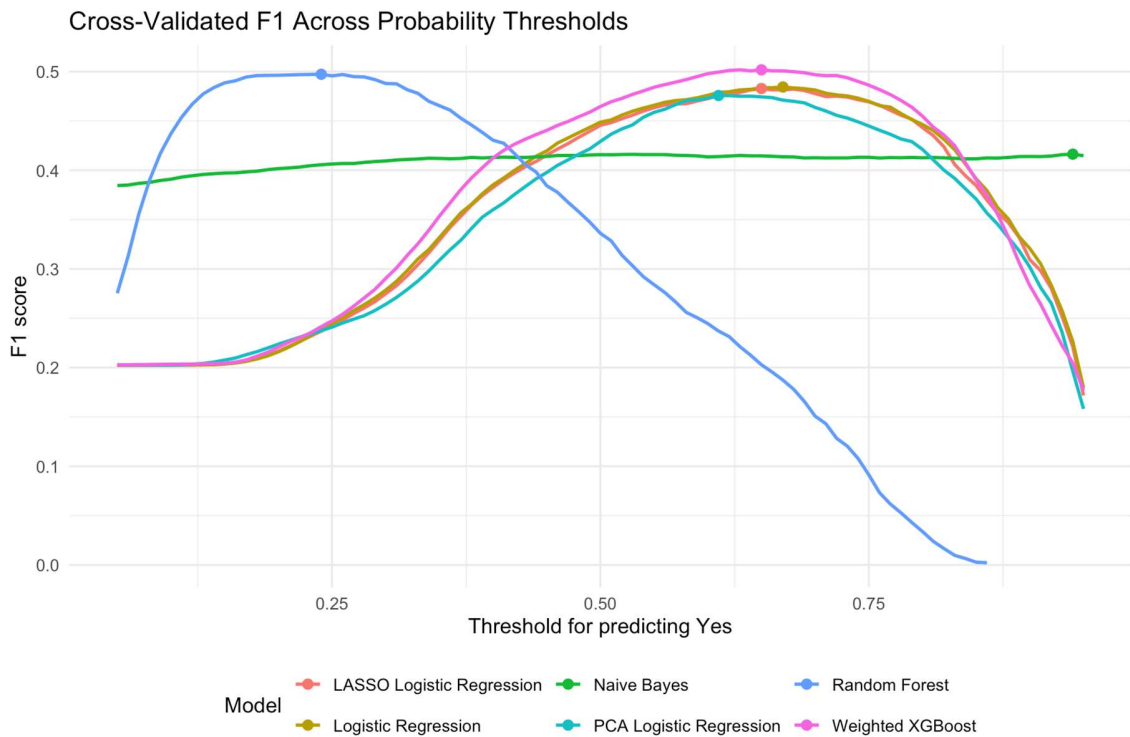


Figure 6. Cross-validated F1 across probability thresholds for each model.

Model	CV PR-AUC	Chosen threshold	CV F1	CV recall	CV precision
Weighted XGBoost	0.4565	0.65	0.5018	0.5730	0.4464
Random Forest	0.4593	0.24	0.4973	0.5579	0.4486
LASSO Logistic	0.4429	0.65	0.4829	0.5655	0.4213
Logistic Regression	0.4416	0.67	0.4844	0.5504	0.4326
PCA Logistic	0.4364	0.66	0.4808	0.5480	0.4283
Naive Bayes	0.3265	0.94	0.4163	0.4838	0.3653

Table 3. Cross-validated model comparison and F1-based threshold selection.

The threshold choices differ substantially across model families, which shows why one universal cutoff would be misleading. Random Forest reached its best F1 at a much lower probability cutoff than the linear models. Naive Bayes required a much higher cutoff, reflecting its different probability scale. Weighted XGBoost and Random Forest were the strongest cross-validated candidates overall, and both were carried forward to the test-set comparison with their own selected thresholds.

Test-Set Evaluation

The held-out test set provides the main performance comparison. The baseline prevalence is only 11.27%, so an all-No classifier would have about 88.73% accuracy with zero recall and zero F1 for the subscription class. The tuned models intentionally trade some raw accuracy for the ability to identify potential subscribers.

Model	Thr.	Acc.	Prec.	Recall	Spec.	F1	ROC-AUC	PR-AUC
Weighted XGBoost	0.65	0.8776	0.4663	0.5970	0.9133	0.5236	0.8092	0.5009
Random Forest	0.24	0.8786	0.4689	0.5851	0.9159	0.5206	0.8048	0.4966

Logistic Regression	0.67	0.8718	0.4468	0.5787	0.9090	0.5042	0.7982	0.4731
LASSO Logistic	0.65	0.8662	0.4314	0.5894	0.9014	0.4982	0.7974	0.4692
PCA Logistic	0.66	0.8700	0.4407	0.5722	0.9078	0.4979	0.7966	0.4596
Naive Bayes	0.94	0.8557	0.3886	0.4903	0.9020	0.4335	0.7627	0.3392

Table 4. Held-out test performance using cross-validated thresholds.

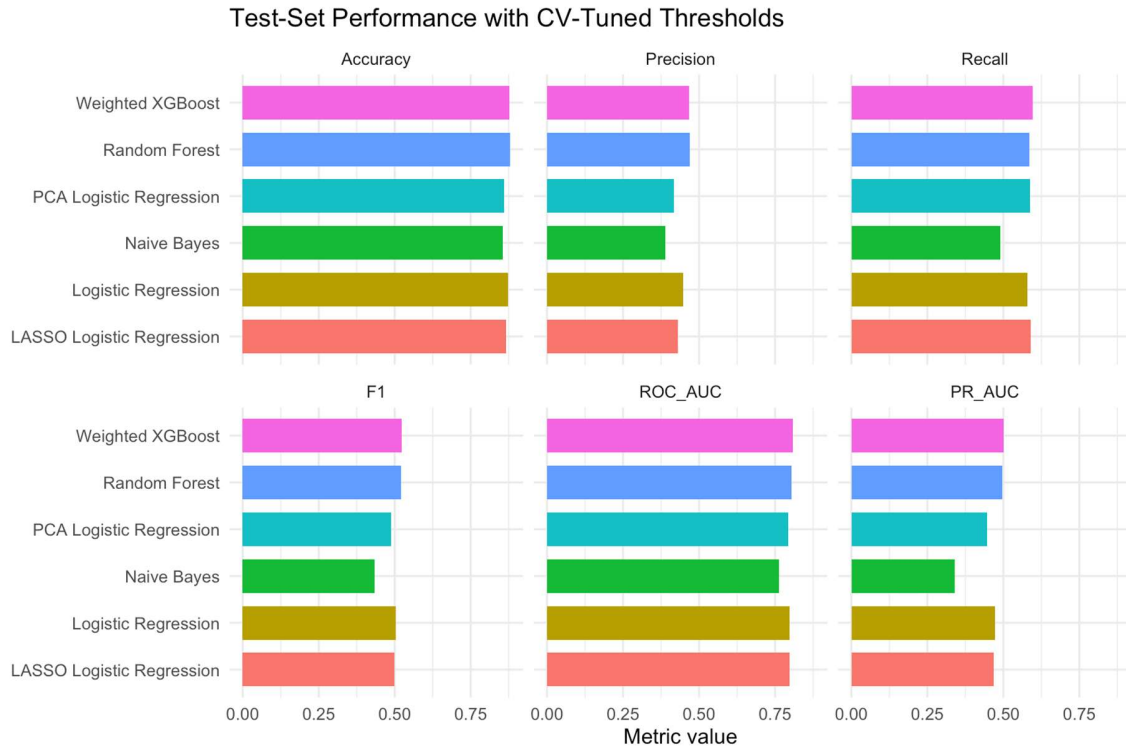


Figure 7. Test-set performance metrics for the cross-validated threshold of each model.

Weighted XGBoost was the best final model by F1, PR-AUC, and ROC-AUC. It identified 554 of the 928 actual subscribers and produced 634 false positive calls. Its advantage came from stronger probability ranking and the ability to model nonlinear interactions among macroeconomic indicators, contact history, and campaign timing. Random Forest was extremely close, with slightly higher precision, specificity, and accuracy. It missed 11 more subscribers than XGBoost, which lowered recall and F1 in a campaign context where recovered positives matter.

The linear models remained competitive, especially given their simpler additive structure. Logistic regression reached a test F1 of 0.5042, and LASSO reached a test F1 of 0.4982. Both were below the ensemble models in PR-AUC, suggesting weaker ranking of rare subscribers. PCA logistic regression remained close to LASSO in F1, but its lower PR-AUC indicates that compressing the numeric predictor block did not improve probability ranking over the original linear feature space. Naive Bayes had the lowest PR-AUC and F1. Its conditional-independence assumption is poorly matched to correlated economic variables and related campaign indicators.

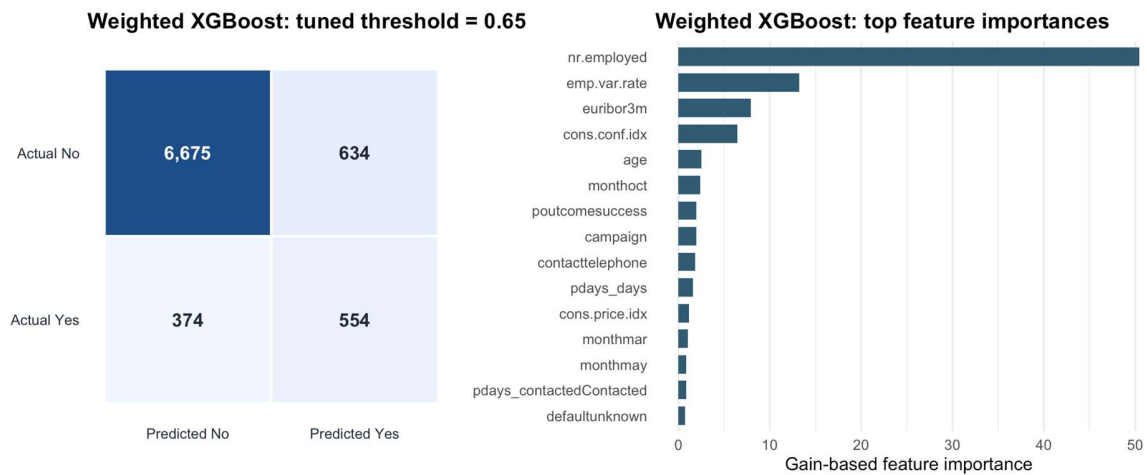


Figure 8. Weighted XGBoost tuned-threshold confusion matrix and gain-based feature importance.

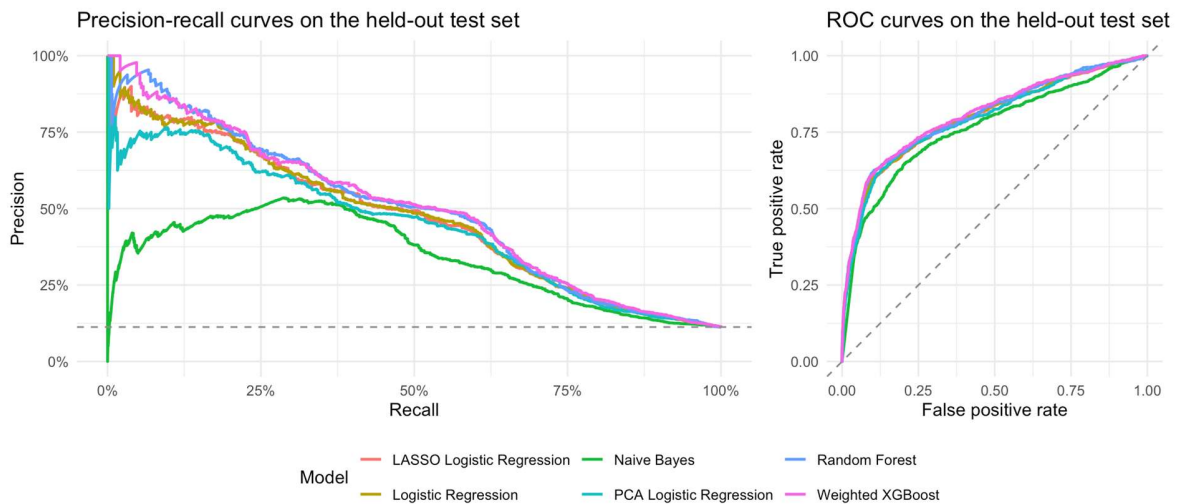


Figure 9. Precision-recall and ROC curves on the held-out test set.

Interpretation and Insights

The XGBoost importance profile in Figure 8 provides the main interpretation for the final model. Importance is measured by gain, so the values describe each feature's contribution to split quality in the boosted trees rather than a signed causal effect. The dominant feature was `nr.employed`, followed by `emp.var.rate`, `euribor3m`, and `cons.conf.idx`. This ranking indicates that the model relied heavily on macroeconomic labor-market conditions, interest-rate context, and consumer sentiment when assigning subscription probability.

Campaign and contact variables formed the next layer of predictive information. The importance of `poutcomesuccess` supports the EDA result that prior successful contact is a strong signal. The roles of `monthoct`, `monthmar`, and `monthmay` show that campaign timing shaped the probability ranking. Campaign count, `contacttelephone`, `pdays_days`, and `pdays_contactedContacted` indicate that contact channel, repeated outreach, and recency of prior contact helped the model distinguish likely subscribers from the majority class. Age appeared in the top features, yet its importance was far smaller than the macroeconomic variables.

From a business perspective, the best model depends on the cost of a call and the value of a successful subscription. If the bank wants to maximize the number of possible subscribers found, Weighted XGBoost is the preferred model, as it produced the highest recall among the two best-performing models and the highest overall F1. If the bank wants to reduce wasted calls more aggressively, Random Forest is a nearly equivalent option due to its slightly higher precision.

Both models are far more useful than accuracy-focused rules, since they recover a meaningful share of the rare positive class.

The PCA benchmark was helpful, yet not dominant. PCA was restricted to the 9 numeric predictors and required 7 components to retain 95% of training-set variance; the resulting model achieved a test F1 of 0.4979. This suggests that compressing the numeric predictor space retains useful information, while preserving categorical predictors separately avoids applying ordinary PCA to one-hot indicators. The result reinforces a common lesson from statistical learning. Dimension reduction can stabilize correlated numeric predictors without automatically improving classification when the response depends on nonlinear interactions or sparse categorical effects.

Conclusion and Reflection

This project demonstrates that careful validation choices matter more than raw accuracy in imbalanced bank marketing prediction. The final recommendation is to use Weighted XGBoost as the primary targeting model, with Random Forest as a strong precision-oriented alternative and logistic/LASSO regression as interpretable benchmarks for explaining directional effects. The strongest predictors point to macroeconomic climate, previous campaign success, previous contact status, contact channel, and month. These findings translate a predictive exercise into operational guidance. A bank should not treat every client as equally likely to subscribe, and it should evaluate targeting models by the quality of positive-class retrieval rather than by overall accuracy.

The analysis has several limitations. First, the train/test split was random and stratified. A future deployment study should use time-based validation, since the dataset is ordered by campaign date, and economic conditions change over time. Second, the report tuned thresholds using F1. In operational use, a bank would choose thresholds using a cost-benefit matrix that assigns values to successful subscriptions and costs to unsuccessful contacts. Finally, probability calibration should be examined before operational deployment, especially since different models required very different decision thresholds.

References

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Appendix A: Reproducible Code

This compact appendix records the reproducible workflow used for the report. The complete executable version is in `bank_marketing_classical_discriminative.Rmd`.

```
set.seed(474)
library(tidyverse)
library(caret)
library(glmnet)
library(e1071)
library(ranger)
library(xgboost)
library(pROC)

bank_raw <- read.csv("bank-additional-full.csv", sep = ";")
bank_model <- bank_raw %>%
  select(-duration) %>%
  mutate(
    pdays_contacted = if_else(pdays == 999, "Not_Contacted", "Contacted"),
    pdays_days = if_else(pdays == 999, 0, pdays),
    y = factor(y, levels = c("no", "yes"), labels = c("No", "Yes")),
    across(where(is.character), as.factor)
  ) %>%
  select(-pdays)

train_id <- createDataPartition(bank_model$y, p = 0.80, list = FALSE)
train_data <- bank_model[train_id, ]
test_data <- bank_model[-train_id, ]
fold_out <- createFolds(train_data$y, k = 5, returnTrain = FALSE)
fold_in <- map(fold_out, ~ setdiff(seq_len(nrow(train_data)), .x))
pos_weight <- sum(train_data$y == "No") / sum(train_data$y == "Yes")
w <- if_else(train_data$y == "Yes", pos_weight, 1)

cv_ctrl <- trainControl(
  method = "cv", number = 5, classProbs = TRUE,
  summaryFunction = bank_summary,
  savePredictions = "final", index = fold_in, indexOut = fold_out
)

fit_glm <- train(y ~ ., train_data, method = "glm", family = binomial(),
  metric = "PR_AUC", weights = w, trControl = cv_ctrl,
  preProcess = c("zv", "center", "scale"))
fit_lasso <- train(y ~ ., train_data, method = "glmnet",
  tuneGrid = expand.grid(alpha = 1, lambda = 10 ^ seq(-4, -0.3, length.out = 35)),
  metric = "PR_AUC", weights = w, trControl = cv_ctrl,
  preProcess = c("zv", "center", "scale"))
fit_nb <- train(y ~ ., train_data, method = e1071_nb_model,
  tuneGrid = expand.grid(laplace = C(0, 1)),
  metric = "PR_AUC", trControl = cv_ctrl,
  preProcess = c("zv", "center", "scale"))
fit_rf <- train(y ~ ., train_data, method = "ranger",
  tuneGrid = expand.grid(mtry = c(4, 8, 12, 18),
    splitrule = "gini", min.node.size = c(10, 25)),
  metric = "PR_AUC", trControl = cv_ctrl,
  num.trees = 400, class.weights = C("No" = 1, "Yes" = pos_weight))

# Weighted XGBoost used the same folds, positive-class weights,
# and PR-AUC ranking over sampled tree settings.
# Thresholds were selected from out-of-fold probabilities by maximum F1.
# Final metrics were computed once on the held-out test set.
```

Appendix B: AI / LLM Usage Documentation

在 UCI 银行营销数据集的二分类预测中，正样本仅占约 11.3%。在这种严重的类别不平衡下，为什么传统的准确率（会是一个具有误导性的指标

核心概念

Accuracy 看的是 **整体预测正确的比例**：

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

但在类别极度不平衡时，它可能看起来很高，却没有实际价值。

为什么有误导性

在 UCI 银行营销数据里：

- positive（订阅）≈ 11.3%
- negative（不订阅）≈ 88.7%

如果模型直接把 **所有客户都预测成“不订阅”**：

- TN 很高
- TP = 0

accuracy 仍然大约是：

88.7%
↓

看起来很好，但银行一个潜在客户都找不到。

Figure B1. Prompt about why accuracy is misleading under 11.3% positive-class imbalance.

核心概念

折外预测（OOF prediction）就是在 k-fold cross validation 中，**每个样本都由“没见过它”的模型预测一次概率**。

为什么这样做

我们想调整 classification threshold（比如从 0.5 调到 0.3/0.6），但：

- 用 training set prediction → 太乐观，因为模型见过数据
- 用 test set → data leakage，因为 test set 应该留到最后

所以用 OOF prediction 最合理。

具体逻辑

比如 5-fold：

- 用 Fold2-5 train → 预测 Fold1
- 用 Fold1,3,4,5 train → 预测 Fold2
- ...

最后每个样本都会得到一个 OOF probability。

然后基于这些 probability：

↓

- lower threshold → higher recall
- higher threshold → higher precision

Figure B2. Prompt about out-of-fold prediction logic for threshold tuning.

核心概念

- **False Positive (FP)**: 预测客户会订阅，但实际上不会。
- **False Negative (FN)**: 预测客户不会订阅，但实际上会。

为什么重要

不同错误对应不同 business cost。

- FP → 联系了错误客户
- FN → 错过了真实潜在客户

实际影响

False Positive:

- 浪费电话营销成本
- 占用销售人员时间
- 可能引起客户反感

False Negative:

- 错失潜在存款收入
- 降低 campaign ROI
- 可能让竞争对手抢走客户



Figure B3. Prompt about false-positive/false-negative business costs.